Machine Learning
Homework 1

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1. In linear regression, we learned to find the optimal parameters by minimizing the loss function. However, when the model is too complex or the feature dimension is too high, overfitting is prone to occur. Regularization is a key technique to solve this problem.

   (a) Briefly explain the core differences between Ridge Regression and Lasso Regression regarding the following two aspects:

   1. Their effect on model weights.
   2. Their suitability for feature selection. Which method is more suitable and why?

   (b) Consider the cost function for Ridge Regression:

   $$J(\boldsymbol{w}, b) = \frac{1}{m}\sum_{i=1}^{m}(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i + b - y_i)^2 + \frac{\lambda}{2}||\boldsymbol{w}||_2^2, \tag{1}$$

   Let $\tilde{\boldsymbol{w}} = \begin{bmatrix}\boldsymbol{w}\\ b\end{bmatrix}$ and $\tilde{\boldsymbol{x}}_i = \begin{bmatrix}\boldsymbol{x}_i\\ 1\end{bmatrix}$ to absorb the bias term into the weight vector. Rewrite the cost function in its matrix form, $J(\tilde{\boldsymbol{w}})$. Then, derive the closed-form solution for $\tilde{\boldsymbol{w}}$ by solving $\nabla_{\tilde{\boldsymbol{w}}} J(\tilde{\boldsymbol{w}}) = 0$. Finally, prove that this solution is unique for any regularization parameter $\lambda > 0$.

Ans:

(a) Ans for (a)

(b) Ans for (b)

2. Consider the binary cross-entropy loss for a single training example (x_i, y_i):

$$J_i(\boldsymbol{\beta}) = -[y_i \log(p_1(\hat{\boldsymbol{x}}_i; \boldsymbol{\beta})) + (1 - y_i)\log(1 - p_1(\hat{\boldsymbol{x}}_i; \boldsymbol{\beta}))], \tag{2}$$

where $\boldsymbol{\beta} = (\boldsymbol{w}; b)$, $\hat{\boldsymbol{x}}_i = (\boldsymbol{x}_i; 1)$, $p_1(\hat{\boldsymbol{x}}_i; \boldsymbol{\beta}) = \sigma(\boldsymbol{\beta}^T\hat{\boldsymbol{x}}_i)$ is the predicted probability that $y_i = 1$, and $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

(a) Derive the gradient of this loss function with respect to the parameter vector $\boldsymbol{\beta}$, which is $\nabla_{\boldsymbol{\beta}} J_i(\boldsymbol{\beta})$. (Hint: You will need to use the property of the sigmoid function's derivative: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.)

(b) Now, let's add an L2 regularization term to the logistic regression cost function. The new cost function for the entire dataset is:

$$J_{\text{reg}}(\boldsymbol{\beta}) = \left(\frac{1}{m} \sum_{i=1}^{m} J_i(\boldsymbol{\beta}) \right) + \frac{\lambda}{2m} ||\boldsymbol{w}||_2^2. \tag{3}$$

Note that regularization is applied only to the weights $\boldsymbol{w}$, not the bias term b. Write down the gradient descent update rules for both the weights $\boldsymbol{w}$ and the bias b based on this regularized cost function $J_{\text{reg}}(\boldsymbol{\beta})$.

Ans:

(a) Ans for (a)

(b) Ans for (b)

3. For the standard multivariate linear regression model, the model is expressed in matrix form as $\mathbf{y} = \mathbf{X}\boldsymbol{w} + \boldsymbol{\epsilon}$. The ordinary least squares solution, which minimizes the residual sum of squares $||\mathbf{y} - \mathbf{X}\boldsymbol{w}||_2^2$, is given by the normal equation: $\boldsymbol{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ (assuming $\mathbf{X}^T\mathbf{X}$ is invertible). The predicted values are $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{w}^*$, and the residual vector is defined as $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.

Prove that the residual vector $\mathbf{e}$ is orthogonal to the column space of the feature matrix $\mathbf{X}$. Mathematically, this means proving that $\mathbf{X}^T\mathbf{e} = \mathbf{0}$.

Ans: Ans for this question.